

The **Online Shoppers Purchasing Intention Dataset (2020)** provided by [Henry Sue on Kaggle](#) is a popular benchmarking tool used to predict whether a website visitor will complete a purchase. It contains data from **12,330 sessions** collected over a one-year period.

Dataset Overview

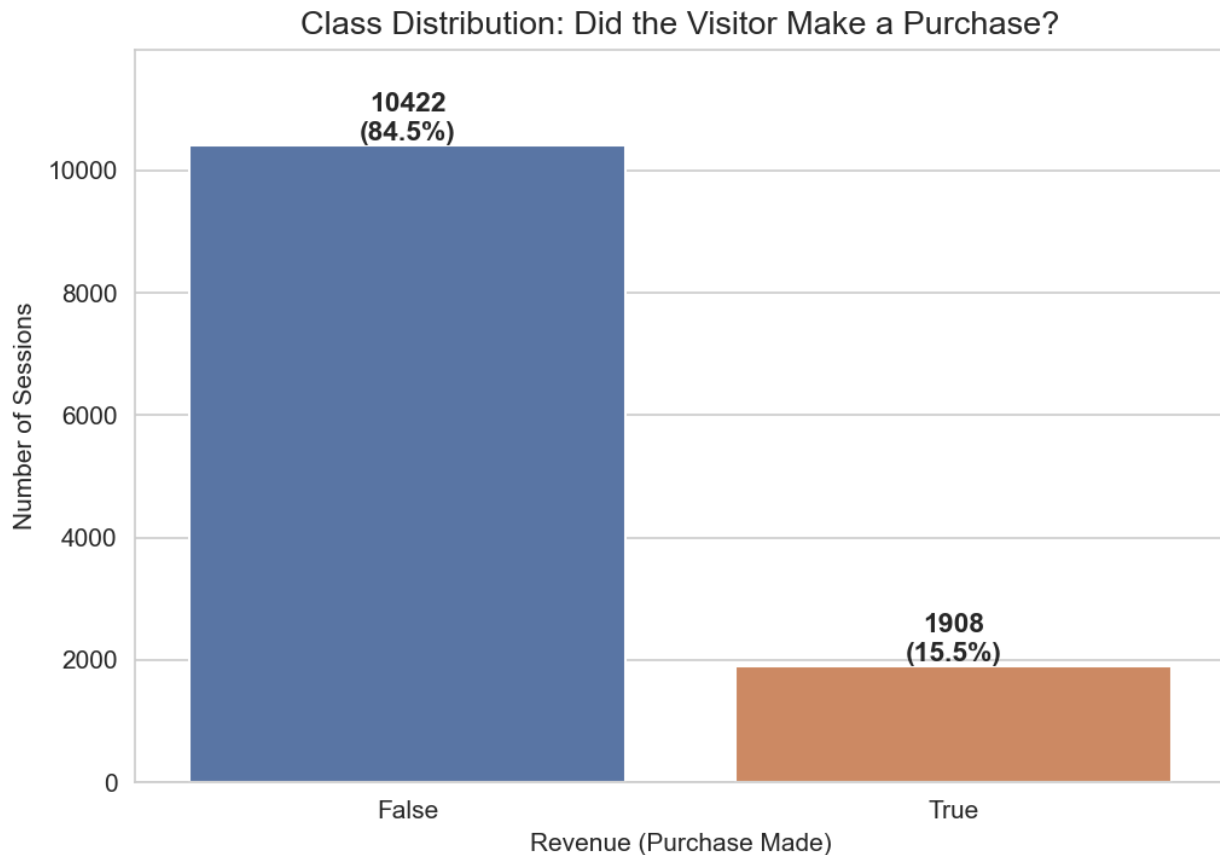
- **Source:** Originally sourced from the [UCI Machine Learning Repository](#).
- **Composition:** 10 numerical and 8 categorical attributes.
- **Class Label:** The Revenue attribute (True/False) indicates if a purchase was made.
- **Imbalance:** The dataset is highly imbalanced, with roughly **84.5%** of sessions resulting in no purchase and only **15.5%** ending in a sale.

Key Attributes & Features

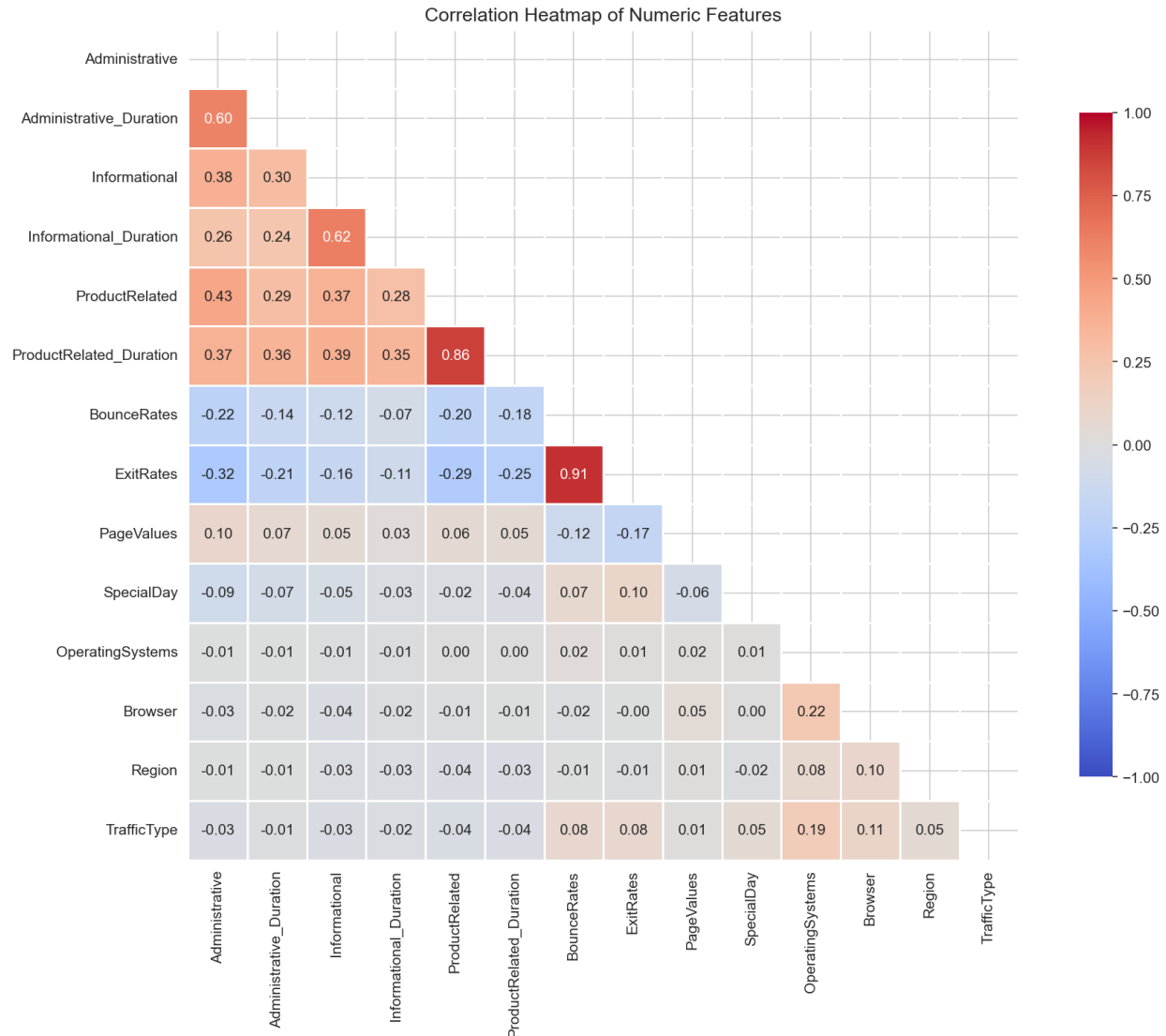
The features are divided into behavioral metrics and session metadata:

Category	Features	Description
Page Visits	Administrative, Informational , ProductRelated	Number of pages visited in each category.
Time Spent	...Duration	Total time spent in each of the page categories mentioned above.
Google Analytics	BounceRate, ExitRate, PageValue	Metrics derived from Google Analytics for each page. PageValue is often the strongest predictor of revenue.
Temporal	Month, OperatingSystems, Browser, Region, TrafficType	General metadata about the user's connection and the time of the visit.
Visitor Info	VisitorType, Weekend	Whether the user is returning or new, and if the visit occurred on a weekend.

FINDINGS ON THE DATA:

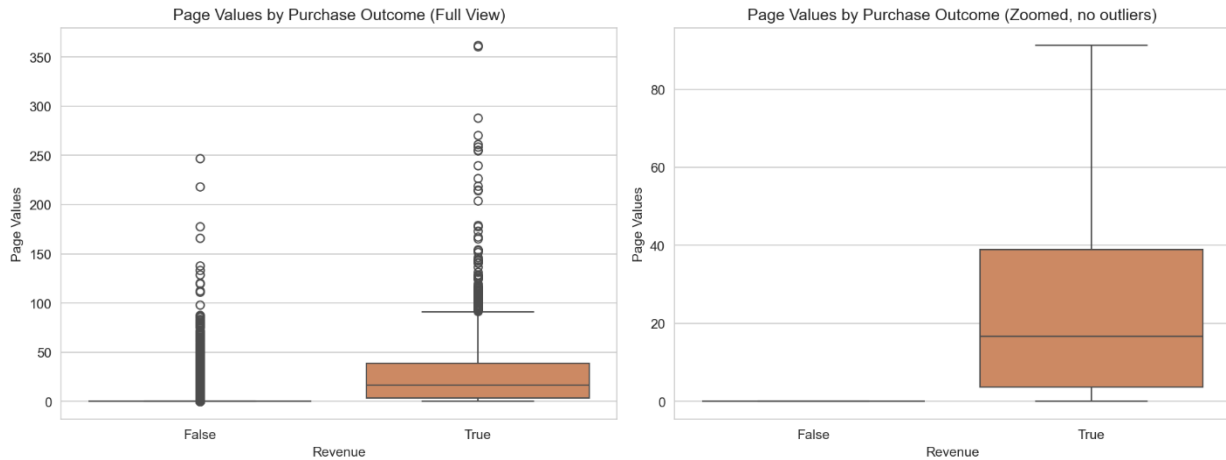


1. For every 1 person who bought something, about **5.5** people just browsed and left. This is normal for e-commerce — most online visitors don't buy. The dataset exhibits significant class imbalance, with only 15.47% of sessions resulting in a *purchase*.
 - The model would be **84.53%** accurate without doing any actual learning, because it would be right every time the visitor didn't buy (which is most of the time).
 - **So accuracy alone is a trap here.** A model can look great on paper while being completely useless. This is why our capstone will report multiple metrics — *accuracy, precision, recall, F1-score*, and the *confusion matrix* — so the real performance shows through.
 - This imbalance necessitates the use of **multiple evaluation metrics** beyond accuracy to ensure meaningful assessment of model performance.



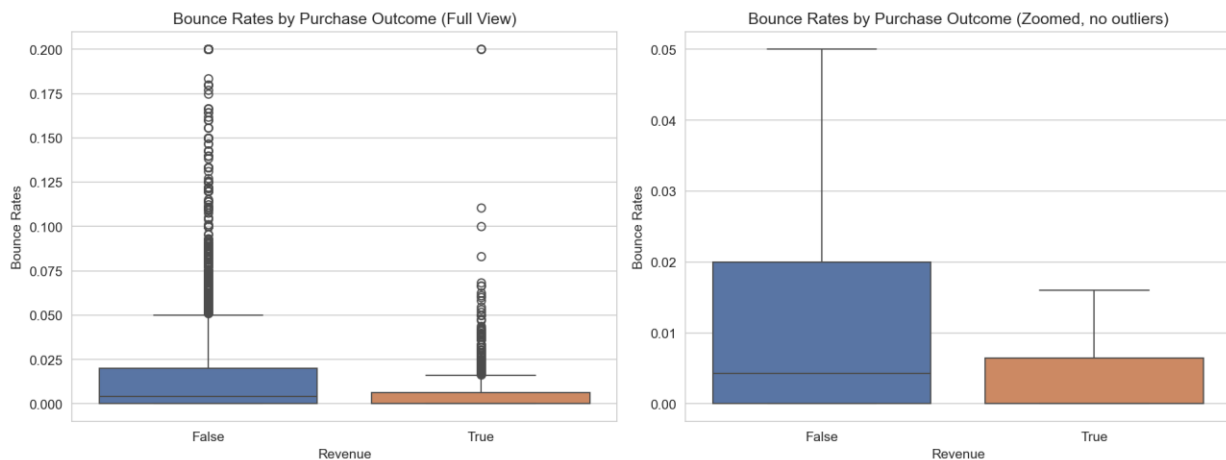
2. Some of the features are basically saying the **same thing**. *BounceRates* and *ExitRates* at **0.913** are **duplicates**. *ProductRelated* and its duration at **0.861** are nearly duplicates too.

- *kNN* measures "**distance**" between visitors using all features at once. When two features are duplicates, they essentially get counted twice in the distance calculation, giving them **double influence**. For example, if Visitor A bounced a lot AND exited a lot, *kNN* counts that behavior twice — once via *BounceRates*, once via *ExitRates*. Not a fatal one, but something honest to mention in our **Limitations** section.



3. This is Google Analytics' measure of how "valuable" the pages someone visited are. A high **Page Value** means they were on pages that typically lead to sales. A *Page Value* of zero means they only saw pages that don't drive sales.

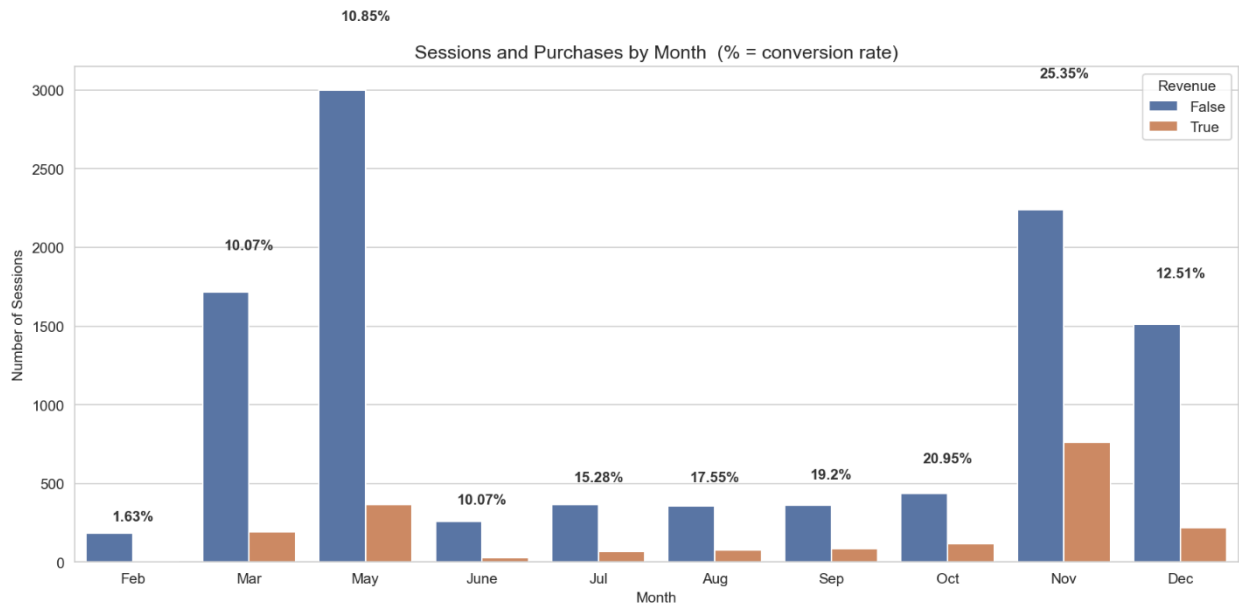
- People who didn't buy: median **Page Value** = 0. People who bought: median **Page Value** = 16.76. The mean for buyers (27.26) is almost **14 times higher** than for non-buyers (1.98). This is what data scientists call **strong class separability** — and it's exactly what *kNN* needs to work well.
- **Page Values** emerged as the most discriminative feature: visitors who completed purchases had a mean **Page Value** of 27.26, compared to only 1.98 for non-purchasers — a **13.8-fold difference**.



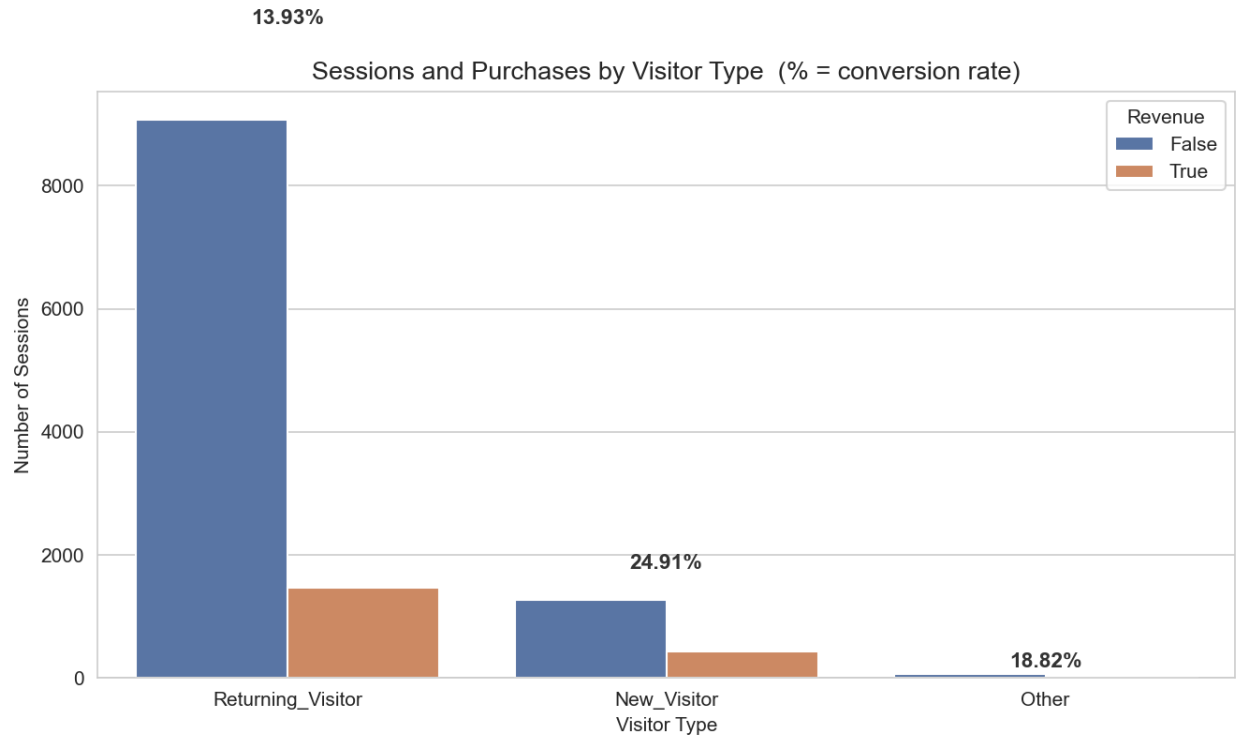
4. The percentage of times a visitor lands on a page and immediately leaves. Higher **bounce rate** = more "drive-by" behavior. Non-buyers had *bounce rates* roughly **5 times higher** than buyers (0.0253 vs 0.0051). The median *bounce rate* for buyers is literally **zero**.

- **Buyers** stay and engage; **non-buyers** drift in and out.

- Buyers showed substantially lower *bounce rates* (mean = **0.0051**) compared to non-buyers (mean = 0.0253), confirming that **engagement depth** is strongly associated with purchase completion.



- Where the traffic is: *May (3,364)* and *November (2,998)* had the most visitors. Where the buyers are: *November* isn't just busy — it's the **highest converting month** at **25.35%**.
 - There's a clear upward trend from *July (15.28%)* to *November (25.35%)*. The model will learn that if a session happens in November, there's a **higher prior probability of purchase**.
 - Conversion rates** varied significantly by month, ranging from **1.63%** in February to **25.35%** in November.



6. **New visitors** convert at **24.91%**, while **returning visitors** convert at only **13.93%**. New visitors buy at almost **2 times the rate** of returning visitors! New visitors often arrive with **high purchase intent**, while returning visitors include lots of "**browsers**."
- This is a perfect **discussion point**. When someone asked "what insights did we find?", we can lead with this **counterintuitive result**. It shows we didn't just plug numbers into a model — we actually ***understood*** what the data was saying.